CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

May 2012

EE 310 Programmable Logic Controller & Industrial Automation

Date: 11.05.2012, Friday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 State and Explain Advantages and Disadvantages of PLC. [07]

Q - 2 (a) Explain major units of a PLC. Also draw PLC system layout. [07]

(b) Explain DC Input Module and AC Input module. [07]

OR

Q - 2 (a) Explain PID module with its block diagram. [05]

(b) Explain 8 bit ADC converter with Digital Step diagram. [04]

(c) Draw and Explain PLC CPU power supply. [05]

Q - 3 (a) Using Ladder diagram make NOT, OR, AND, NAND, NOR, EX-OR and EX-NOR Gate. [09]

(b) What are the different types of Input and Output circuits? Explain each. [05]

OR

Q - 3 (a) Describe TON, TP and TOFF blocks available in Codesys V2.3 [06]

(b) State the D-Morgan's theorem. And prove it by making ladder logic in PLC. [08]

SECTION – II

Q - 4 Draw and explain PLC Process scanning in detail. [07]

Q - 5 (a) Write down the program for the following logic: [10]

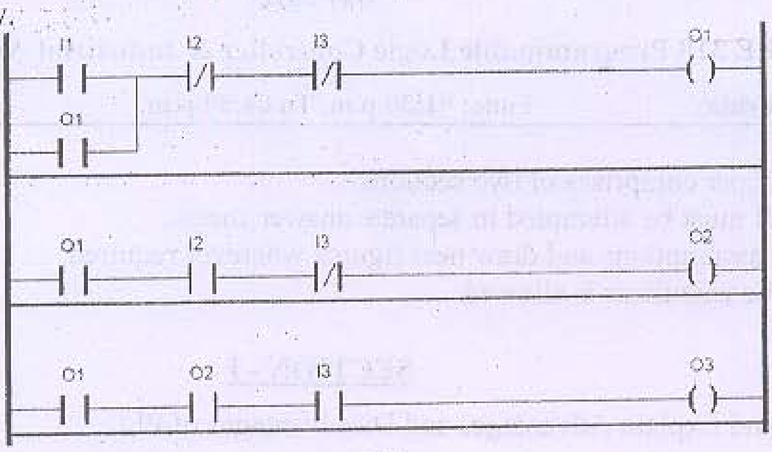
A wood saw W, a fan F and a Lube pump P, all go on when a start button is pushed. A stop button stops the saw only. The fan is to run an additional 5 seconds to blow the chips away. The lube pump is to run for 8 seconds after shutdown of W. Additionally, if the saw has run more than one minute, the fan should stay on indefinitely. The fan may then be turned off by pushing a separate fan reset button. If the saw has run less than one minute, the pump should go off when the saw is turned off. The 8 second time delay off does not take place for a running time of less than 1 minute.

(b) Write down the program for the following logic: [04]

There are three mixing devices on a processing line: A, B and C. After process begins, mixer A is to start after 7 seconds elapse. Next, mixer B is to start 3.6 seconds after mixer A. Mixer C is to start 5 seconds after B. All then remain on until a master enable switch is turned off.

OR

Q - 5 (a) For the Given ladder logic diagram write down the output (O1, O2 and O3) is TRUE or FALSE for the given input (I1, I2 or I3) condition. All Logic should be done in sequence only. **[06]**



Seq. No.	Input Conditions			Output coils		
				O1	O2	O3
1	I1: TRUE	I2: FALSE	I3: FALSE			
2	I1: FALSE	I2: FALSE	I3: FALSE			
3	I1: FALSE	I2:TRUE	I3: FALSE			
4	I1: FALSE	I2:TRUE	I3: TRUE			

(b) A Thermo-couple is connected to a 16 bit Analog Input module. Maximum input voltage of Analog input module is 24 V DC. Assume that for 0^0 C to 200^0 C thermo-couple gives 0 V to 20 V DC linearly. At output digital module 5 nos. of Fans are connected. So write down the program for the following given logic. **[08]**

When temperature is (i) $\geq 0^0$ C but $<40^0$ C, One fan is on. (ii) $\geq 40^0$ C but $<80^0$ C, two fans are on (iii) $\geq 80^0$ C but $<120^0$ C, three fans are on (iv) $\geq 120^0$ C but $<160^0$ C, four fans are on (v) $\geq 160^0$ C all fans are on.

Q - 6 (a) Write Down a program using Timer block (TON, TOF or TP) for the following bulb series as shown in Figure 1. **[08]**

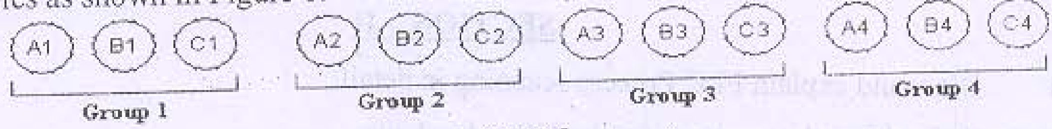


Figure 1

Group 1 on for 150ms (at that time all other groups are off), after that group 2 is on for next 150ms (at that time all other groups are off), after that group 3 is on for next 150ms (at that time all other groups are off), after that group 4 is on for next 150ms (at that time all other groups are off). After this all process is repeated until processor is on.

(b) What are the factors to be considered in selecting a PLC? Describe each in short. **[06]**

OR

Q - 6 (a) Explain CTU, CTD and CTUD function block. Draw the timing diagram of CTU. **[08]**

(b) Explain Memory map organization with necessary diagram. **[06]**

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